

1. In the rocket thought experiment, why does Einstein connect gravity with acceleration?
2. A neutron star has a mass of 2.8 solar masses and a radius of 12 km. If 30 s pass far from the neutron star, how much time passes on its surface?
3. GPS satellites orbit 2.66×10^7 m from Earth's center, from the circular orbits page. Their clocks are farther from Earth's gravity than clocks on the surface (6.37×10^6 m from the center). Over one day of far-away time (86 400 s), how many more microseconds tick on a GPS clock than on a surface clock? Use Earth mass 5.97×10^{24} kg, and ignore the satellite's speed.

4. Why does gravity bend the path of light if photons have no rest mass?

5. Earth has mass 5.97×10^{24} kg. If Earth were compressed until it became a black hole, what would its Schwarzschild radius be?

6. In 2015, LIGO detected gravitational waves from two black holes merging into one black hole of about 62 solar masses. What is the Schwarzschild radius of the final black hole?

7. A neutron star has a mass of 2.0 solar masses and a radius of 12 km. Is it a black hole?

8. The event horizon is sometimes drawn as a dark surface. Is it a solid surface that an astronaut would hit?
9. Astronomers estimate a black hole event horizon radius of 45 km. Assuming it is non-rotating, what mass does the Schwarzschild radius equation give?
10. A white dwarf star has a radius of 7000 km. If 1.0 hour passes far from the white dwarf, how much time passes on its surface?
11. A clock hovers 60 km from the center of a non-rotating 10 solar mass black hole. Use 1 solar mass = 2.0×10^{30} kg. If 2.0 s pass far from the black hole, how much time passes on the hovering clock?

12. Why does general relativity predict gravitational waves?

13. A non-rotating black hole has mass 10 solar masses. How far from its center would a hovering clock need to be for its time to run at half the far-away rate?

14. In 2019, the Event Horizon Telescope took the first picture of a black hole, at the center of the galaxy M87. It has a mass of about 6.5 billion solar masses. What is its Schwarzschild radius? Compare it to the distance from Earth to the Sun, 1.5×10^{11} m.

15. Why does a full black hole model probably need both general relativity and quantum mechanics?